

1 Dinamical Systems

1.1 Models of Biological Systems

Logistic Growth Model (Population): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K})$ N: population at time t; K: carrying capacity (depends on resources); r, K: > 0.

Sol: $N(t) = \frac{N_0 K e^{rt}}{K + N_0(e^{rt} - 1)} \rightarrow K$ Equilibrium points: $N = 0$ (unstable), $N = K$ (stable). Quickest: $N = \frac{K}{2}$ (<凹; >凸)

Delay Model: Model: $\frac{dN}{dt} = f(N(t), N(t - T))$ T: delay time > 0 周期解 | 与初始值无关 ps: To consider e.g. gestation period, maturation time etc.

¹Ext. of Logistic Model: $\frac{dN}{dt} = rN(t) [1 - \frac{N(t-T)}{K}]$ (用"ave. pop." 估计, 准用"Convolution Type") $\Rightarrow$ ²Dimensionless: $\frac{dN^*}{dt^*} = N^*(t^*) [1 - N^*(t^* - T^*)]$ ($N^* = \frac{N}{K}$, $t^* = rt$, $T^* = rT$)

^{1,2} Periodic sol: stable limit cycle $N(t + t_p) = N(t)$, t_p : period. Amplitude: rT Equilibrium Points: $N = 0 | N^* = 0$ (unstable); $N = K | N^* = 1$ (Stable: $0 < T^* = rT < \frac{\pi}{2}$; Unstable: $T^* > \frac{\pi}{2}$)

Continuous Population Model (EN): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - EN = f(N)$ r, K, E: > 0; EN: harvesting rate/unit time; E: effort measure.

Equilibrium: $N = 0$ (unstable); $N = N_h(E) = K(1 - \frac{E}{r}) > 0$ (need $E < r$, Stable) Yield: $Y(E) = EN \cdot E > r$: die out; else: Max Yield: $Y_M = \frac{rK}{4}$ ($E = \frac{r}{2}$) Steady State: $N_h | Y_M = \frac{K}{2}$

Continuous Population Model (Y₀): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - Y_0 = f(N; r, K, Y_0)$ Y₀: constant yield rate/unit time.

Max Yield: $Y_M = \frac{rK}{4}$ Equilibrium: $0 < Y_0 < Y_M$: Two positive points (unstable(小) and stable(大)). Sensitivity: 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

Examples of $N_{t+1} = f(N_t)$: Verhulst: $N_{t+1} = rN_t(1 - \frac{N_t}{K})$; $r, K > 0$. Better: Ricker curve: $N_{t+1} = N_t e^{r(1 - \frac{N_t}{K})}$

Imperfect Bifurcation: $\dot{x} = rx - x^3 + h$ (codimension-2 bifurcation) If $r < 0$, x^* unique. If $r > 0$, $|h| > h_c$, x^* unique; $|h| < h_c$, x^* three. (两边 stable, 中 unstable). Bifurcation points: $h = \pm h_c = \pm \frac{2r}{3} \sqrt{\frac{r}{3}}$. cusp point: $(r, h) = (0, 0)$

1.2 Summary - Stability and Bifurcation

	$\frac{dN}{dt} = f(N)$ No Limit cycle periodic solutions				S : stable; U : unstable.				$N_{t+1} = N_t F(N_t) = f(N_t)$ Eigenvalue: $\lambda = f'(N^*)$							
Stability	Stable: $f'(N^*) < 0$			Unstable: $f'(N^*) > 0$			Stable: $ \lambda < 1$		Unstable: $ \lambda > 1$ (chaotic)		Periodic: $ \lambda = 1$ $\begin{cases} + : \text{tangent bifurcation} \\ - : \text{period-doubling bifurcation} \end{cases}$					
Linearise	$N(t) = N^* + n(t), n(t) \ll 1 \Rightarrow \frac{dn}{dt} = \frac{dN}{dt} = \begin{cases} \text{(general)} = f(N^*) + f'(N^*)n(t) \\ \text{(delay)} = \text{直接带入 } t \text{ 后近似} \end{cases}$							Let $N_t = N^* + n_t, n_t \ll 1 \Rightarrow N^* + n_{t+1} = \dots = f(N^* + n_t) \approx f(N^*) + f'(N^*)n_t \Rightarrow n_{t+1} = \lambda n_t$								
Bifurcation \\\ Perturbations at N^* Periodic	Saddle-Note (Normal Form) $\dot{x} = r \pm x^2$ $0 \rightarrow S \rightarrow U$		Transcritical $\dot{x} = rx - x^2$ $S \rightarrow U \rightarrow S$		(super-)Pitchfork $\dot{x} = rx - x^3$ $S \rightarrow S \rightarrow S$		Sub-critical Pitchfork $\dot{x} = r + x^3$ $U \rightarrow U \rightarrow S$		$\lambda < -1$: $\uparrow$ oscillations		$-1 < \lambda < 0$: $\downarrow$ oscillations		$0 < \lambda < 1$: $\downarrow$ Monotone		$\lambda > 1$: $\uparrow$ Monotone	
									If $f^p(N^*) = N^*$, N^* is p -periodic. $\forall$ even p -periodic N^* with $\lambda = -1$ branches into $2p$ -periodic							
Other	Time-Independent: At N^* : $\frac{dN}{dt} = 0$ $N(t) = N(t - T) = N^*$ $f(N^*) = 0$								Sarkovskii's Theorem: If a 3-periodic N^* exists, then $\forall p \geq 1$, p -periodic N^* exists. Species extinction: $N_{\min} < 1$							
Other	Tangency Condition: If $\dot{x} = f(x, r)$, 分岔点 (x^*, r_c) : $\frac{\partial f}{\partial x}(x^*, r_c) = 0$ $f(x^*, r_c) = 0$ 两点合并: coalesce.								Max/Min: (如果单峰 (unimodal)) 2-periodic $f'(N_m) = 0$, $N_{\max} = f(N_m)$; $N_{\min} = f(N_{\max})$							

Hopf Bifurcation: 平衡点从稳定变为不稳定, 同时产生 limit cycle. 条件: $\det(J) > 0$; $\frac{d\text{tr}(J)}{d\mu} \neq 0$ or: $\text{tr}(J)$ +/- 符号有变化 $\Leftrightarrow \lambda = \pm i\omega$ and $\frac{d\Re(\lambda)}{d\mu} \neq 0$ μ : parameter, not time.

$\Rightarrow$ Stable $\rightarrow$ Unstable + Stable Limit Cycle (supercritical) or Unstable $\rightarrow$ Stable + Unstable Limit Cycle (subcritical). (一般 super-) when $\text{tr}(J) = 0 | \Re(\lambda) = 0$, $\lambda = \pm i\omega \Rightarrow$ periodic: $T = \frac{2\pi}{\omega}$

Bifurcation: Qualitative changes in dynamics are called bifurcations. (当平衡点性质 | 个数发生变化时) Bifurcation Diagram: y 轴: 平衡点; x 轴: 控制参数; 虚线 unstable, 实线 stable.

稳定性 Stability 判别 | For Continuous Model: Find perturbation: $n(t)$ and $\frac{dn}{dt} \Rightarrow$ Assume $n(t) = e^{\lambda t} | ce^{\lambda t} \rightarrow$ Find $\lambda = g(\lambda) \Rightarrow$ Let $\lambda = u + i\omega \rightarrow u = g_1(u)$ and $\omega = g_2(\omega) \Rightarrow u < 0$ asy.stable; $u > 0$ unstable. Find T_c : (如要寻找参数 (e.g. T) 范围/临界值 使得平衡点稳定) Check $T = 0$ stable. $\Rightarrow$ Using $u = 0$ (i.e. $\lambda = i\omega$) to find T_c .

2 Interacting Populations

Lotka-Volterra Model (Predator-Prey): $\frac{dN}{dt} = N(a - bP)$; $\frac{dP}{dt} = P(cN - d) \Rightarrow$ nondim: $\frac{dN}{d\tau} = u(1 - v)$; $\frac{dP}{d\tau} = \alpha v(u - 1)$ Equilibrium: $(u^*, v^*) = (0, 0), (1, 1)$

Exact sol: $\alpha u + v - \ln(u^\alpha v) = H$, $H > H_{\min} = \alpha + 1$ constant. (closed phase) Conservation Law: Using $\frac{du}{dv} = \frac{du}{d\tau} \frac{d\tau}{dv}$ find $H \Rightarrow \frac{dH}{d\tau} = 0$, here it satisfies. (By chain rule)

Show Level curves (constant of motion) closed: If $H(u, v)$ is level curve, Look $\begin{bmatrix} H_{uu} & H_{uv} \\ H_{vu} & H_{vv} \end{bmatrix}_{(u^*, v^*)} \Rightarrow$ If +/- definite (i.e. $\forall \lambda_i > 0 (< 0)$) $\Rightarrow$ It's local min/max at (u^*, v^*) & strictly convex/concave $\Rightarrow$ closed curves. Closed curves $\Rightarrow$ periodic sol.

Determinant Stability: For model: $\begin{bmatrix} u'(t) \\ v'(t) \end{bmatrix} = \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix}$ at equilibrium (u^*, v^*) . Let $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u - u^* \\ v - v^* \end{bmatrix}$, then $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} f_u & f_v \\ g_u & g_v \end{bmatrix}_{(u^*, v^*)} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow$ Find eigenvalues λ of J by $\det(J - \lambda I) = 0$

¹ $\det(J) > 0$, $\text{tr}(J) < 0 \Rightarrow \Re(\lambda) < 0 \rightarrow$ stable node $\text{Im}(\lambda) = 0$; stable focus=spiral point $\text{Im}(\lambda) \neq 0$; ² $\det(J) > 0$, $\text{tr}(J) > 0 \Rightarrow \Re(\lambda) > 0 \rightarrow$ unstable node $\text{Im}(\lambda) = 0$; unstable focus=spiral point $\text{Im}(\lambda) \neq 0$

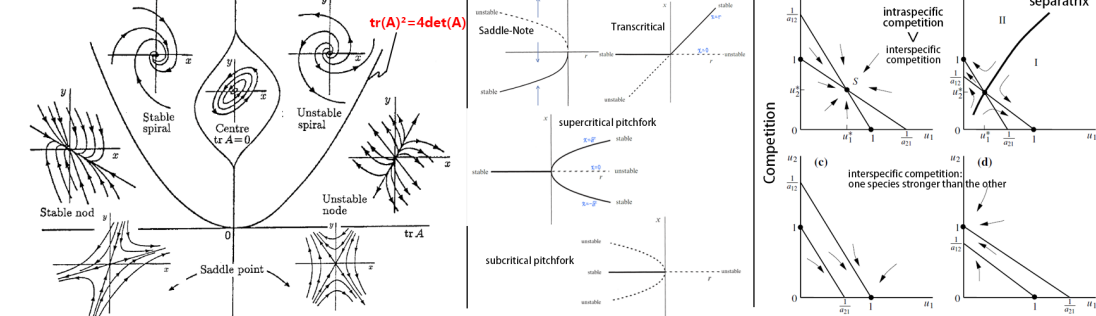
³ $\det(J) < 0 \Rightarrow \text{Im}(\lambda) = 0$, $\lambda_1 < 0 < \lambda_2 \rightarrow$ saddle point (unstable); ⁴ $\det(J) > 0$, $\text{tr}(J) = 0 \Rightarrow \lambda = \pm i\omega$ (pure imaginary) $\rightarrow$ center (only linear stability) | Hopf bifurcation (见前面)

Nullclines: $f(u, v) = 0$ or $g(u, v) = 0 \rightarrow$ 得 u, v 表达式是 nullclines. Trapping Region: A closed region R 可以用直线/null clines 构成 s.t. 所有 边界的外法线方向 $\mathbf{n}$, 在对应边界上的点, 满足 $\mathbf{n} \cdot \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix} < 0$

Find slope of Nullclines: $f(u, v) = 0 \Rightarrow df = f_u du + f_v dv = 0 \Rightarrow \frac{dv}{du} = -\frac{f_u}{f_v}$ Similarly: $\frac{dv}{du} = -\frac{g_u}{g_v}$

Poincare-Bendixson Theorem: One unstable node/focus/spiral point + it's in trapping region $\Rightarrow$ A stable limit cycle exists. (Cannot be a saddle (since $\det(J)$ cannot be < 0))

Competition Model: principal of competitive exclusion: If two species compete for the same limited resource, one will eventually outcompete and exclude the other. (b, c, d)



3 Reaction Kinetics

Enzyme Kinetics: e.g. $S + E \xrightleftharpoons[k_{-1}]{k_1} SE \xrightarrow{k_2} P + E$ K_i : rate of reaction; S: substrate; E: enzyme; SE: complex; P: product. Rate of uptake: $v = \frac{d[P]}{dt} |_{t=0}$

Law of Mass Action: Rate of reaction is proportional to the product of the concentrations of the reactants. e.g. $\alpha_1 A + \alpha_2 B \xrightleftharpoons[k_{-1}]{k_1} \beta_1 C + \beta_2 D \Rightarrow$ rateforward $= k_1 [A]^{\alpha_1} [B]^{\alpha_2}$; ratebackward $= k_{-1} [C]^{\beta_1} [D]^{\beta_2}$

Quasi-Steady State Approximation (QSSA): 如果系统里有快和慢变量, 则可以假设快变量达到稳态, 即 $\frac{d(\text{fast})}{dt} = 0$. (一般说来假设 complex concentration changes rapidly)

Singular Perturbations Analysis: For $\frac{du}{d\tau} = f(u, v)$, $\varepsilon \frac{dv}{d\tau} = g(u, v)$, $0 < \varepsilon \ll 1 \Rightarrow v$ fast (when $g \neq 0$), u slow. Quasi-steady-state approximation: $\varepsilon \rightarrow 0 \Rightarrow g(u, v) = 0$

Taylor: $u(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n u_n(\tau)$; $v(\tau, \varepsilon) = \sum_{n=0}^{\infty} \varepsilon^n v_n(\tau)$ Outer solution: Taylor 带入原方程组 $\rightarrow$ 保留 $\mathcal{O}(1) | \varepsilon = 0$ Valid for $\tau > 0$ (slow time); $v_0(0)$ invalid. $\rightarrow$ not uniformly valid for $\tau \geq 0$

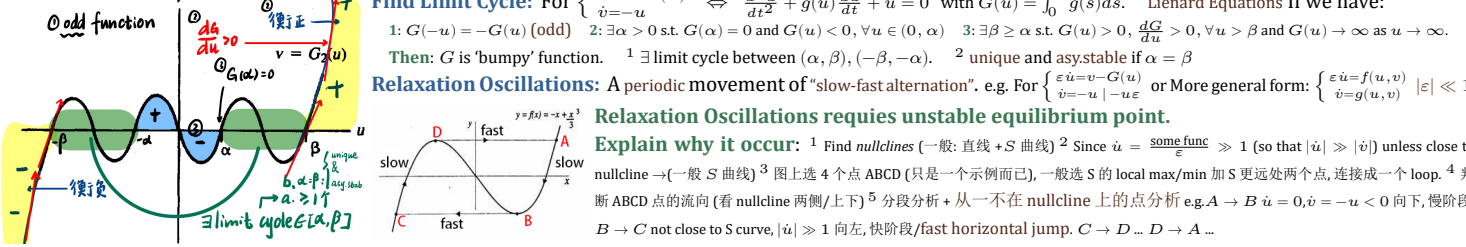
Inner solution: $\sigma = \frac{\tau}{\varepsilon}$, $V(\sigma; \varepsilon) = v(\tau; \varepsilon) \Rightarrow \frac{dV}{d\sigma} = \varepsilon f(u, v)$, $\frac{dV}{d\sigma} = g(u, v) \rightarrow$ Taylor 带入原方程组 $\rightarrow$ 保留 $\mathcal{O}(1) | \varepsilon = 0$ (known as boundary layer ($\tau = 0$ 处)); Valid $\tau = 0$ (fast time)

Asymptotic Matching: $\lim_{\sigma \rightarrow \infty} [U_0(\sigma), V_0(\sigma)] = \lim_{\tau \rightarrow 0^+} [u_0(\tau), v_0(\tau)]$ Uniformly Valid Solution: $u(\tau; \varepsilon) = u_0(\tau) + \mathcal{O}(\varepsilon)$; $v(\tau; \varepsilon) = \begin{cases} v_0(\tau) + \mathcal{O}(\varepsilon), & 0 < \varepsilon \ll \tau \\ V_0(\sigma) + \mathcal{O}(\varepsilon), & 0 < \tau \ll 1 \end{cases}$ uniform expansion: 见 chp.5

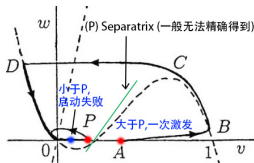
Activator-Inhibitor Model: For $\frac{du}{dt} = f(u, v)$, $\frac{dv}{dt} = g(u, v)$: If $\frac{\partial g}{\partial u} > 0$, u is an activator of v ; If $\frac{\partial f}{\partial v} < 0$, v is an inhibitor of u . ps: Possible have bifurcation behaviour

Positive/Negative Feedback: At Equilibrium: Positive: $f_u, f_v \geq 0$; $g_u, g_v \leq 0$ or $f_u, f_v \leq 0$; $g_u, g_v \geq 0$ Negative: $f_u, f_v \geq 0$; $g_u, g_v \geq 0$ or $f_u, f_v \leq 0$; $g_u, g_v \leq 0$

4 Relaxation Oscillations & Switch Behaviour



Estimate period T : 例如上图 (例子忽略了快反应) $T = \int_{t_A}^{t_B} dt + \int_{t_C}^{t_D} dt = 2 \int_{t_A}^{t_B} dt$ Find $dt = -du$ 带入计算 e.g. In slow time: $\dot{u} = 0 \rightarrow v = G(u)$ and $\dot{v} = -u \rightarrow \frac{dv}{dt} = \frac{dv}{du} \frac{du}{dt} = G'(u) \frac{du}{dt} = -u \rightarrow -\frac{G'(u)}{u} du = dt$, 带入计算



Excitable System: FitzHugh-Nagumo Model $\begin{cases} \frac{dx}{dt} = c[y + x - \frac{x^3}{3} + z(t)] \\ \frac{dy}{dt} = -x + a + by \end{cases}$ (x : excitability variable; y : recovery variable; $z(t)$: external stimulus)

Threshold 现象: **Small perturbation:** return to equilibrium (failed initiation/excitation). **Large perturbation:** large excursion before returning to equili. (action potential|successful initiation). After excitation| 大扰动之后: Refractory Period 不应期: system is unresponsive to further stimulation. **constant stimulus:** periodic firing (limit cycle).

Reason: 寻找 $\dot{u}, \dot{v}$ s.t. $\ll, \gg 1$ i.e. $|\dot{u}| \gg |\dot{v}|$ or $|\dot{u}| \ll |\dot{v}|$

Switch Behaviour: 从一个 stable point 到另一个 stable point. (前面 excitable system|relaxation oscillation 如有两 stable points 且画图分析 (像 relaxation oscillation 那样分析) 可得到)

描述话语 example: 快-慢震荡: ¹ (option) The equilibrium point S is unstable between the two knees/turning points of the u -nullcline/cubic-like curve. | or: S changes stability as it crosses the knees of the u -nullcline. ² 图上标/简单解释拐点位置. ³ 证明 $|u'| \gg 1$ (写成乘积形式判断) 或其他 $\rightarrow$ 得快运动边. ⁴ Trajectories starting off the u -nullcline rapidly/quickly approach left/right branch of the u -nullcline, and then slowly move down/up/crawl along the u -nullcline due to $v' < 0 / > 0$, and to upper/lower knee/turning point ⁵ At the knee/turnning point, the trajectory quickly jumps to the right/left branch of the u -nullcline, and then slowly moves up/down/crawl along the u -nullcline due to $v' > 0 / < 0$. ⁶ And this process repeats periodically, giving rise to a relaxation oscillation. **Excitable:** 类似的, 分析 small perturbation above/below equilibrium point $S \rightarrow$ 得 failed excitation or successful excitation.

5 Manifolds & Fenichel's Theorem | SIR Model

Regular/Singular Perturbation: 对于一个含 ε 的方程/系统: If $\varepsilon \rightarrow 0$ the order does not change, it's a **regular perturbation**. If $\varepsilon \rightarrow 0$ the order changes, it's a **singular perturbation**.

Find Roots: Assume roots $x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$, 代入比较 $\mathcal{O}(1), \mathcal{O}(\varepsilon), \dots$ 得系数(解前两系数就行). If singular, cannot approx $x = \mathcal{O}(\varepsilon^{-1})$, rescaling $x = \frac{X}{\varepsilon}$, assume root $X = X_0 + \varepsilon X_1 + \dots$ 类似前. 不留 $X_0 = 0$ 解, 因为 X 小.

Sol & Uniform: 对 $\varepsilon \dot{y} + p(x)\dot{y} + q(x)y = r(x)$, a. 类似上面算 sol (有初值) or b. 直接算 exact sol. $\rightarrow$ 得 **Outer Solution** (首项含 ε). Let $\tau = \varepsilon t$ 类似算 sol $\rightarrow$ **Inner Solution** (首项不含 ε). **unif expr** = Outer + Inner - Matching(=matching cond of the lim)

Singular Perturbation System: For $\begin{cases} \dot{u} = f(u, v, \varepsilon) \\ \dot{v} = \varepsilon g(u, v, \varepsilon) \end{cases}, 0 < \varepsilon \ll 1. \quad \text{If } \tau = \varepsilon t, \text{ then } \begin{cases} \varepsilon u' = f(u, v, \varepsilon) \\ v' = g(u, v, \varepsilon) \end{cases} \quad \begin{matrix} t: \text{fast time}; \tau: \text{slow time} \\ (\varepsilon \neq 0 \text{ 时等价}) \end{matrix} \quad \downarrow \text{Defined in a neighborhood of } \mathcal{M}_0$

Reduced Problem: slow times $\varepsilon = 0$. **Layer Problem:** fast time $\varepsilon = 0$. **Critical Manifold:** $\mathcal{M}_0 := \{(u, v) : f(u, v, 0) = 0\}$ ps: Reduced problem defined only on $\mathcal{M}_0$, Layer problem defined off $\mathcal{M}_0$

Normal Hyperbolicity: $\mathcal{M}_0$ is normally hyperbolic at $(u, v) \in \mathcal{M}_0$ if: λ_i (特征值) of Jacobian $J|_{(u, v, 0)}$ are $\frac{\partial f}{\partial u}|_{(u, v, 0)}$ are $\Re(\lambda_i) \neq 0$. $\rightarrow$ **(Un-)Stable:** $\Re(\lambda_i) < 0 (> 0) \Leftrightarrow \frac{\partial f}{\partial u} < 0 (> 0)$ (1-dim)

Outer Solution: sol for slow time at $\varepsilon = 0$. (reduced problem) **Inner Solution:** sol for fast time at $\varepsilon = 0$. (layer problem) $\uparrow$ attracting(stable) / repelling(unstable)

Describe Reduced|Layer Problem: ¹ 如果能解具体表达式就解; 不好解情况下给出 u', v' 仅关于 u, v 的表达式; ² 如果可以画一个图; ³ 分析 attracting/repelling (stable/unstable); ⁴ 分析 $\tau, t \rightarrow \infty$ 时的行为

Fenichel's Theorem: If $\mathcal{M}_0 \subset \{f(u, v, 0) = 0\}$ is compact (closed & bounded) and normally hyperbolic. Suppose f, g are smooth. Then for $\varepsilon > 0$ (sufficiently small), $\exists$ **Manifold** $\mathcal{M}_\varepsilon$ s.t. $\mathcal{O}(\varepsilon)$ close and diffeomorphic to $\mathcal{M}_0$. (i.e. $\mathcal{M}_\varepsilon$ is locally invariant under the flow of the full system (上面左边的 system)). ps: $\mathcal{M}_\varepsilon$ called **slow manifold**.

Find $\mathcal{M}_\varepsilon$: If $\mathcal{M}_0 = \{(u, v) : u = p_0(v)\}$, then $\mathcal{M}_\varepsilon = \{(u, v) : u = p_\varepsilon(v)\}$ 可通过算 u, v 的 exact sol, 寻找能使含 ε 项消失但保留 τ 的解 (or: 考虑快变量 $t \rightarrow \infty$, 对比 $\mathcal{M}_0$ 得到 $\mathcal{M}_\varepsilon$ $\downarrow$ **Infective Period:** $= \frac{1}{\alpha}$)

SIR: S: # people who are susceptible. (易感人群) I: # people who are infected. (感染者) R: # people who have been infected and recovered. (no longer susceptible). (康复者) **Total Pop:** $N = S + I + R$

Model: $S' = -\beta SI, I' = \beta SI - \alpha I, R' = \alpha I, S, I \geq 0$. **Assumptions:** ¹ Mass Action incidence: βN transmit infection/unit time; ² αI recover/unit time; ³ No im/emigration, births, deaths. ⁴ No deaths from disease. $\rightarrow$ 新感人数 $= \beta SI$ rate.

临界值: if $S > \frac{\alpha}{\beta}, I \uparrow$ 疾病爆发; If $S < \frac{\alpha}{\beta}, I \downarrow$ 疾病消退. $\rightarrow$ **Basic Reproduction Number:** (A threshold quantity) $R_0 = \frac{\beta S_0}{\alpha}$ or $\frac{\beta N}{\alpha}$, ¹ $R_0 > 1$ epidemic occurs; ² $R_0 < 1$ disease dies out.

$\cdot I_\infty = \lim_{t \rightarrow \infty} I(t) = 0; S_\infty = (S + I)_\infty > 0; N = S_0 + I_0$ **Final Size Relation:** $\ln(\frac{S_0}{S_\infty}) = \frac{\beta}{\alpha} [N - S_\infty] = R_0 [1 - \frac{S_\infty}{N}]$ As $(S + I)' = -\alpha I \rightarrow N - S_\infty = \alpha \int_0^\infty I dt$; As $\frac{S'}{S} = -\beta I \rightarrow \ln(\frac{S_0}{S_\infty}) = \beta \int_0^\infty I dt$

Attack Rate/Ratio: $N - S_\infty$ (# people ever infected | 总感染人数 | Final size of epidemic) **Orbits Sol:** $I(t) + S(t) - \frac{\alpha}{\beta} \log(S(t)) = N - \frac{\alpha}{\beta} \log(S_0)$ **Peak:** $I_{\max}$ when $S = \frac{\alpha}{\beta} \leftrightarrow I' = 0$ (代入前面得精确 $I_{\max}$)

6 Appendix

$y' + p(x)y = q(x)$: **Integrating Factor:** $I(x) = e^{\int p(x)dx} \Rightarrow I(x)y = \int I(x)q(x)dx$ **Taylor Expansion:** $f(x, r)$ at (x_0, r_0) : $f(x, r) = f(x_0, r_0) + (x - x_0)f_x|_{(x_0, r_0)} + (r - r_0)f_r|_{(x_0, r_0)} + \frac{1}{2} \dots$

Binomial Series: $(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \dots$ **Linerisation:** $f(x + \Delta x) = f(x) + f'(x)\Delta x + \dots$ $e^{bx} = 1 + bx + \frac{(bx)^2}{2!} + \dots$ $\ln(a + x) = \ln(a) + \frac{x}{a} - \frac{x^2}{2a^2} + \frac{x^3}{3a^3} - \dots$

Hyperbolic Functions: $\sinh x = \frac{e^x - e^{-x}}{2}$ $\cosh x = \frac{e^x + e^{-x}}{2}$ $\tanh x = \frac{\sinh x}{\cosh x}$ $\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$ $\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$ $\tanh(x) = x - \frac{x^3}{3} + \frac{2x^5}{15} - \dots$